



Thermal, structural and electrical study of boron-incorporated LATP glasses and glass-ceramics



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ABSTRACT

Glasses of three compositions were prepared in the $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{B}_2\text{O}_3-\text{TiO}_2-\text{P}_2\text{O}_5$ system. Next, they were heat-treated at 900°C for 2 h to yield fast Li^+ ion conducting glass-ceramics of NASICON-like structure. Effects of boron additive on structural, thermal and electrical properties of glasses and glass-ceramics are herein investigated and discussed. Thermal analysis revealed that glass transition and crystallization temperatures shift towards lower values with increasing boron content. Moreover, the system under study exhibits improved glass-forming ability. X-ray diffraction studies show that B-incorporated glass-ceramics are mainly composed of rhombohedral $R\bar{3}c$ phase with unit cell parameters a and c decreasing with boron substitution. Raman peaks at 791 cm^{-1} and 1206 cm^{-1} in glasses with B_2O_3 addition confirm the presence of BO_4 and BO_3 units in the glass structure. Overall ion transport properties (total conductivity and activation energy) were characterized by means of impedance spectroscopy. For optimized composition, the highest total conductivity of $6 \cdot 10^{-5}\text{ S/cm}$ for B-incorporated glass-ceramics was recorded. Moreover, separation of the bulk and grain boundary electrical responses was performed on the basis of equivalent circuit modelling of impedance data.

1. Introduction

Li^+ conducting crystalline and glassy materials have been intensively studied with a view of replacing currently used liquid electrolytes in the next generation of portable power sources, the so-called all-solid-state batteries. It is believed that they might offer higher energy density and improved safety than currently used lithium-ion batteries, but their commercialization is hindered by several important factors. Key issue is still insufficient ambient temperature conductivity from 10^{-5} to 10^{-3} S/cm , while 10^{-2} S/cm is rather required [1].

Much work has been carried out on the lithium ion conductors with NASICON-type rhombohedral $R\bar{3}c$ structure of general formula $\text{LiM}_2(\text{PO}_4)_3$, where $\text{M} = \text{Ti}, \text{Ge}, \text{Hf}, \text{Zr}, \text{Sn}$ [2]. The compound $\text{Li}_{1+x}\text{Al}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ (denoted as LATP) with $x = 0.3$ presents the most favourable characteristics for fast ionic migration [3], which results in the highest bulk ionic conductivity among all oxide-based inorganic lithium electrolytes reaching $3.4 \cdot 10^{-3}\text{ S/cm}$ [4]. However, the total electrical properties are heavily impeded by high grain boundary resistance of 10^{-5} up to 10^{-6} S/cm [5,6].

LATP ceramics have been synthesised using several techniques, e.g. solid state reaction followed by sintering [7,4,3], ball-milling [8,9], sol-gel [10–12], co-precipitation [13] or crystallization of glass [14,15]. The latter approach, glass-ceramic route, requires preparing the glassy/

amorphous analogue of intended composition and submitting it to controlled crystallization at a moderate temperature. In comparison to standard sintering method, the precipitation of LATP phase from glass is reputed to be responsible for further conductivity increase due to the porosity reduction and homogeneous microstructure which could be effectively controlled by appropriate heat treatment [16].

Several studies investigating the crystallization process in the $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{TiO}_2-\text{P}_2\text{O}_5$ system have shown that with a careful crystallization control, a maximum conductivity of 10^{-3} S/cm can be obtained [17,18]. However, the precipitation of the main LATP phase is often corrupted by the formation of undesirable phases which hinder the migration of Li^+ ions. To overcome this inconvenience, H. Chen et al. [19] proposed to add to the melt boron oxide, B_2O_3 , a good glass former with a low melting point. They concluded that this modification improved the glass-forming tendency and promoted the precipitation of the LATP phase leading to a high conductivity value of $1.3 \cdot 10^{-3}\text{ S/cm}$ for a glass $14\text{Li}_2\text{O} \cdot 6\text{Al}_2\text{O}_3 \cdot 3\text{B}_2\text{O}_3 \cdot 38\text{TiO}_2 \cdot 39\text{P}_2\text{O}_5$ crystallized at 900°C for 12 h.

The present work relates to the processing as well as thermal, structural and electrical characterization of glasses and glass-ceramics in the $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{B}_2\text{O}_3-\text{TiO}_2-\text{P}_2\text{O}_5$ system. Three compositions of precursor glasses were chosen in which Al^{3+} content was fixed at 0.3 mol% while B^{3+} content varied from 0 to 0.3 mol%. By that means

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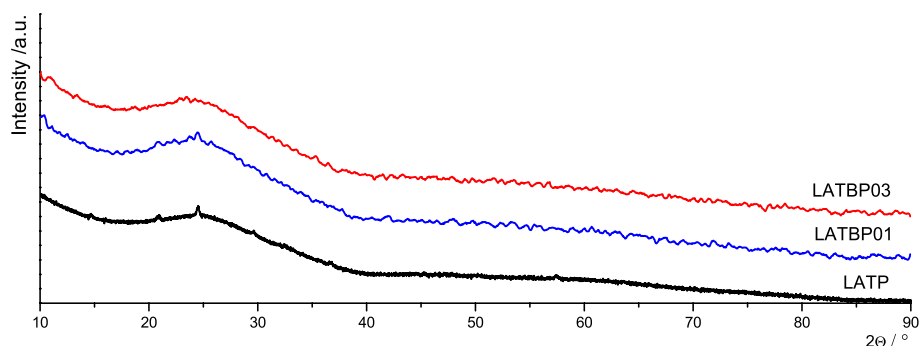


Fig. 1. XRD patterns of as-cast LATP, LATBP01 and LATBP03 glasses.

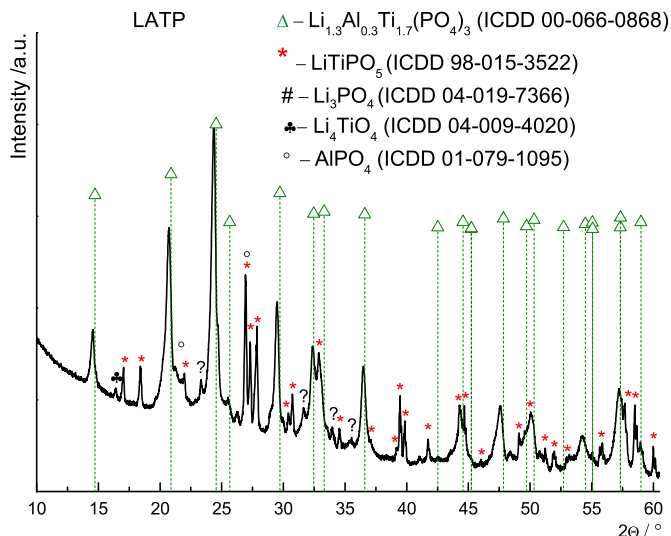


Fig. 2. XRD pattern of the LATP glass-ceramics obtained via crystallization at 900 °C for 2 h along with identified crystalline phases.

B³⁺ ions were expected to substitute for Ti⁴⁺ in the NASICON-like structure and the risk of unwanted crystallization of secondary, boron-containing phases should be minimized.

2. Experimental

2.1. Material processing

Three batches of nominal compositions (mol%): 16.5Li₂O · 3.75Al₂O₃ · 42.5TiO₂ · 37.25P₂O₅ (LATP glass), 17.5Li₂O · 3.75Al₂O₃ · 1.25B₂O₃ · 40TiO₂ · 37.5P₂O₅ (LATBP01 glass) and 20Li₂O · 3.75Al₂O₃ · 3.75B₂O₃ · 35TiO₂ · 37.5P₂O₅ (LATBP03 glass) were prepared using reagent grade chemicals: Li₂CO₃ (AVANTOR), Al₂O₃ (MERCK), H₃BO₃ (POCH), TiO₂ (MERCK) and NH₄H₂PO₄ (AVANTOR). The reagents were weighed, mixed and ground for 30 min in an agate mortar and pestle. Mixed reagents were placed in alumina crucibles and transferred to a graphite crucible of the induction furnace Argenta AFI-02 heated to 1450 °C. They were held at that temperature for 25 min. Next, the melt was quenched between two stainless steel plates. The glassy batches of ca. 5 g were obtained and immediately submitted to annealing at 500 °C for 4 h in order to release internal stresses. Finally, the batches were crystallized and sintered at 900 °C for 2 h. As a result, dense, white specimens were obtained for further characterization.

2.2. Measurements

Glasses and glass-ceramics were characterized by X-ray powder diffraction technique using PANalytical Empyrean diffractometer with

CuKα = 1.542 Å line, set in the Bragg-Brentano geometry operated at 40 kV and 35 mA current with a copper target. The measurements were done at room temperature in air flux. The diffraction peaks positions were measured in the 2θ range from 10° to 90° in 0.016° step with 200 s counting for each step. Phase composition analysis was performed by means of ICDD (International Centre for Diffraction Data) PDF4+ database.

Differential thermal analysis was carried out on Q600 calorimeter (TA Instruments) in the temperature range from 40 °C to 750 °C temperature range at a heating rate of 10 °C/min. An empty alumina pan was used as the reference. The cell was purged with argon during measurement.

Bulk glasses and glass-ceramic materials were manually polished with SiC abrasive paper to remove the spoiled and uneven surface layer. Next, Pt electrodes were sputtered on opposite sides of the samples. The obtained samples were ca. 1 mm in thickness and their surface area was in the range from 8 to 16 mm². The impedance measurements were conducted on computer controlled Solartron1260 impedance/gain-phase analyser in the frequency range from 1 Hz up to 7 MHz and in the temperature range from +30 to +140 °C. At each temperature the specimens were equilibrated for 1 h before the measurement.

The Raman scattering measurements were carried out using Renishaw micro-Raman inVia Reflex spectrometer. The Raman measurements were conducted at room temperature at ambient conditions. The Raman spectra were collected in backscattering geometry using the 514 nm line of an Ar-ion laser. The laser beam was focused through the objective 50× to the spot about 1.5 μm in diameter and was less than 2.0 mW in power. The low laser power was chosen to avoid laser-induced heating effects.

3. Results and discussion

3.1. Amorphousness, phase composition and structure

As shown in Fig. 1, X-ray diffraction patterns for as-cast glasses exhibit a broad halo, characteristic for the short range atom ordering. Although all starting materials were amorphous, after crystallization the glass-ceramics phase composition varied with the boron content. In the case of LATP (see Fig. 2) after heat treatment at 900 °C for 2 h, the most prominent diffraction maxima were ascribed to following phases: Li_{1.3}Al_{0.3}Ti_{1.7}(PO₄)₃ (ICDD 00-066-0868), LiTiPO₅ (ICDD 98-015-3522) and AlPO₄ (ICDD 01-079-1095). LATBP01 and LATBP03 glass-ceramics (Fig. 3) were more phase pure as all major peaks were identified belonging to NASICON-like phase. Only few low-intensity lines (e.g. at 18.5° and 21.6°) were ascribed to minor secondary phases.

The obtained powder profiles for B-incorporated glass-ceramics were refined via le Bail procedure using Jana2006 program [20]. A rhombohedral model (space group R3c) was used with unit cell parameters: *a* = 8.4956 Å, *b* = 8.4956 Å, *c* = 20.8195 Å, α = 90°, β = 90°, γ = 120° provided in literature by Wang et al. [21]. Refinement lattice parameters are given in Table 1. With the increase of boron content a

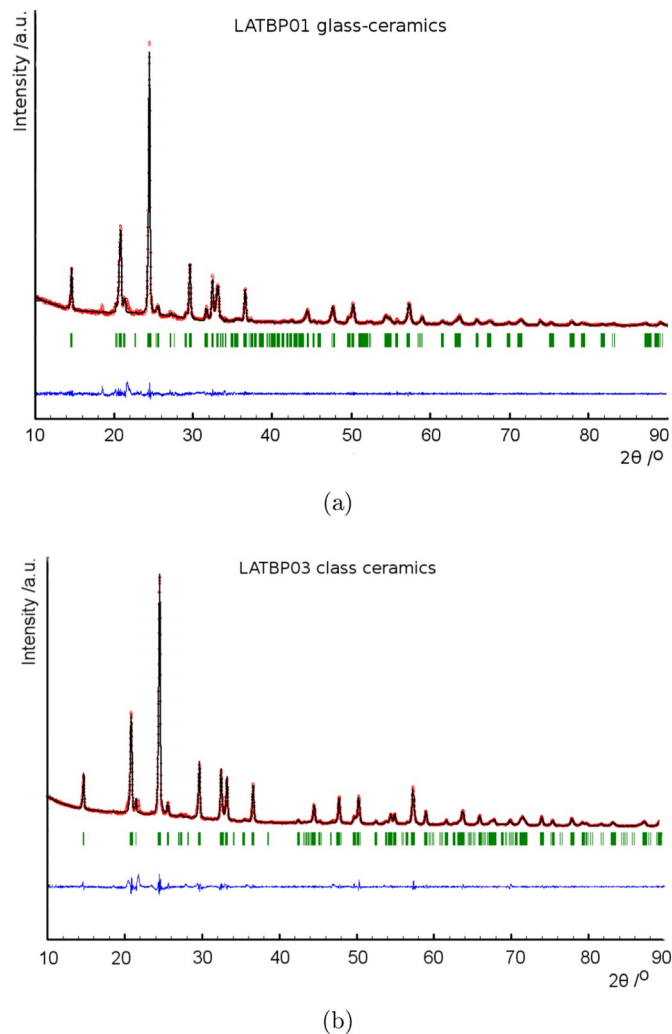


Fig. 3. Diffraction profiles of powdered samples LATBP01 (a) and LATBP03 (b) glass-ceramics fitted using Le Bail algorithm. Observed (○) and calculated (line) profiles are shown, as well as difference profile (lower) with reflection positions indicated by markers.

Table 1

Refinement lattice parameters for studied samples.

Sample	a (Å)	b (Å)	c (Å)	α (°)	β (°)	γ (°)
LATBP01 (this work)	8.52	8.53	21.22	89.56	91.82	120.26
LATBP03 (this work)	8.44	8.53	21.04	90.90	90.06	120.02

lattice constant decreases from 8.52 Å for LATBP01 to 8.44 Å for LATBP03, and c - from 21.22 Å to 21.04 Å, respectively. Moreover, the unit cell becomes more distorted and loses its rhombohedral $R\bar{3}c$ symmetry as evidenced by α , β and γ variations.

The structure of as-cast glasses was examined using Raman spectroscopy method. The collected spectra are shown in Fig. 4 for comparison. LATP glass exhibits the major bands centred at: 982, 868, 635, 497 and 288 cm^{-1} . For glasses containing B_2O_3 additive, there is a new band arising at 791 cm^{-1} , whereas the one at 868 cm^{-1} is suppressed. In addition, a pronounced shoulder appears at 1206 cm^{-1} . The arising peaks at 791 cm^{-1} and 1206 cm^{-1} are related in the literature to the B–O stretching modes of tetrahedral (BO_4) and trigonal (BO_3) units, respectively [22]. This confirms the incorporating of the B_2O_3 oxide in the LATP glass structure.

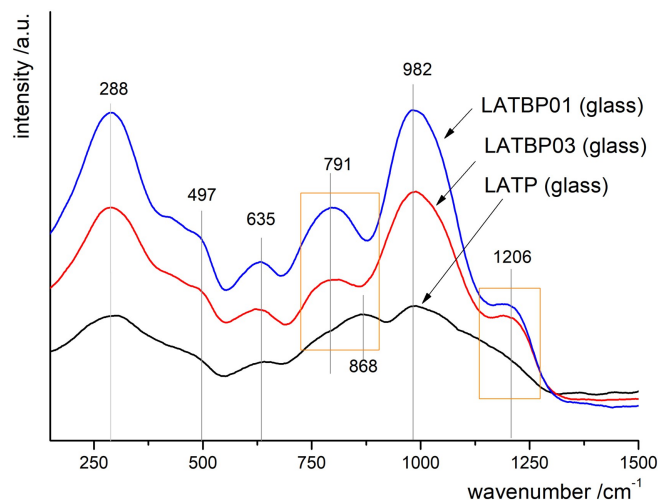


Fig. 4. Raman spectra of LATP, LATBP01 and LATBP03 glasses.

According to literature, Raman spectra of glasses in the system $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{TiO}_2-\text{P}_2\text{O}_5$ are dominated by vibrational features due to phosphate ions and the coordination numbers of Ti^{4+} and Al^{3+} ions seem to vary, facilitating the formation of different types of phosphate species [23]. Bands at 288 and 497 cm^{-1} may be attributed to PO_2

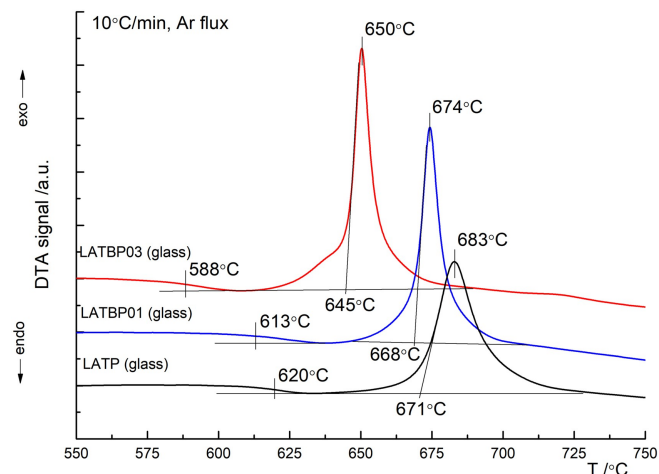


Fig. 5. DTA thermograms of the as-cast glasses at a heating rate of 10 °C/min.

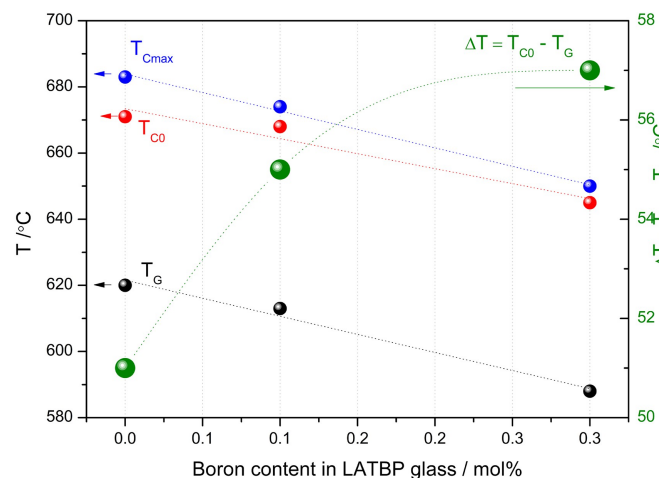


Fig. 6. Glass transition T_g , onset T_{c0} and peak crystallization T_{cmax} temperature as a function of boron content in LATBP materials.

(O–P–O) bending motions, while those observed between 1200 and 900 cm^{-1} – to the symmetrical and asymmetrical vibrations of the PO_4 [24]. The band at 868 cm^{-1} may correspond to the vibrations of TiO_5 polyhedra [25]. A band near 635 cm^{-1} according to [26] is attributed to the vibration of octahedral titanate units TiO_6 .

3.2. Thermal properties

Fig. 5 presents DTA thermograms of the as-cast glasses. In each case the first weak endothermic step-shaped signal corresponds to the glass transition occurring at $T_g = 620^\circ\text{C}$ for LATP, 613°C for LATBP01 and 588°C for LATBP03. Next, crystallization process starts at $T_{c0} = 671^\circ\text{C}$, 668°C , 645°C and reaches its maximum at $T_{cmax} = 683^\circ\text{C}$, 674°C and 650°C for LATP, LATBP01 and LATBP03, respectively. T_g , T_{c0} , T_{cmax} decrease almost linearly with boron content (Fig. 6). ΔT values (difference between onset of crystallization and glass transition temperatures) are relatively small, 50–58 $^\circ\text{C}$, meaning that the system under study exhibits rather poor glass-forming ability. Nevertheless, the

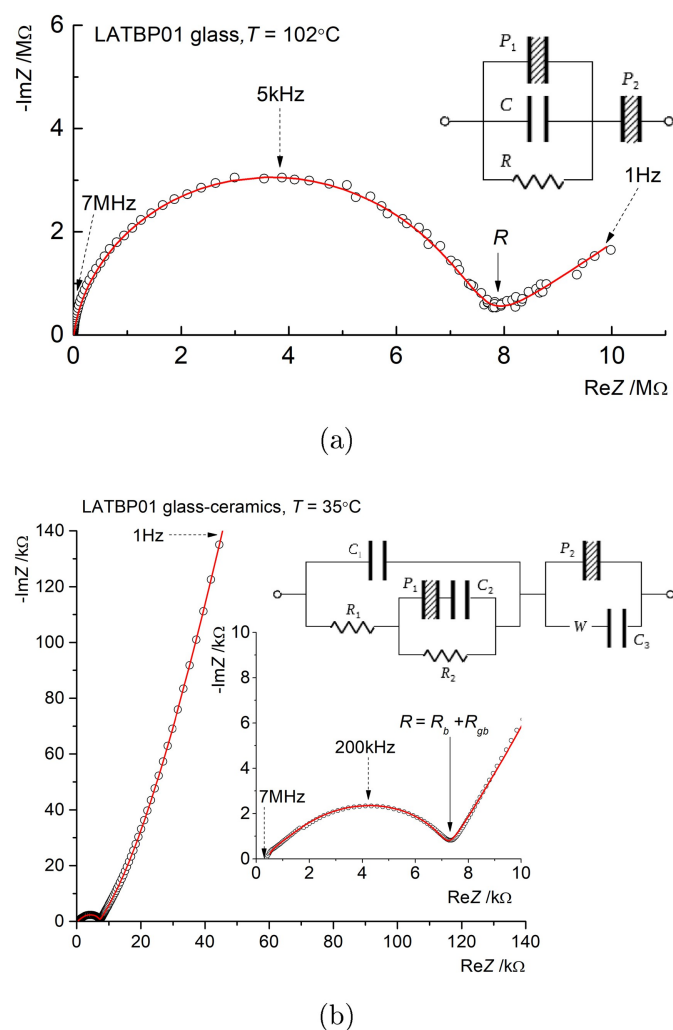


Fig. 7. Complex impedance plots of LATBP glass (a) at 102°C and LATBP01 glass-ceramics at 35°C (b); symbols $\circ$ represent experimental data and solid line is a result of fitting procedure using the equivalent circuit (C - capacitors, P - constant phase elements, R - resistors and W - Warburg element) presented as an inset (for (a): P_1 and C correspond to overall relaxation processes in the bulk; R represents the DC resistance of the sample; P_2 - double-layer formation at the electrodes; for (b): C_1 - geometrical capacitance of the sample; R_1 and R_2 - grain and grain boundary resistances respectively; C_2 and P_1 - electrical polarization occurring on phase boundaries; P_2 , W (Warburg diffusion) and C_3 - electrode part of the circuit).

increase of ΔT (Fig. 6) with boron content suggests that its presence enhances the glass-forming tendency of the B-incorporated LATP-based glasses.

3.3. Electrical properties

The impedance spectroscopy method was used to investigate the electrical properties of LATP, LATBP01 and LATBP03 glasses and glass-ceramics. The complex impedance spectra for LATBP01 as-cast glass (Fig. 7a) consists of one depressed and slightly distorted semicircle representing bulk properties of the sample and an inclined line arising as a result of polarization phenomena at the blocking electrodes. After crystallization, Nyquist plot for LATBP01 glass-ceramics could be separated into three parts - two overlapping semicircles and a straight line. In the high-frequency domain, the first semicircle models the electrical response from grain interior, the other - from grain boundary region and a line in a low frequency range is a result of ion blocking at the Pt electrodes.

The R values marked at Fig. 7 were assumed to be the samples' total DC resistance (bulk for glasses and bulk & grain boundary for glass-ceramics). On that basis, the total ionic conductivity was calculated as $\sigma = d/(R \cdot S)$ (d - sample thickness ca. 1 mm, S - electrode area from 8 to 16 mm^2 depending on the sample). For all glasses under study, the $\sigma(T)$ dependence (Fig. 8) followed the Arrhenius formula with activation energies: 0.55 eV for LATP and 0.63 eV for LATBP01 and LATBP03 (Table 2). The room temperature conductivity (at 25°C), σ_{RT} , for LATP glass was very low (10^{-8} S/cm). For LATBP glasses, the σ_{RT} was even one order of magnitude lower although for these compositions the concentration of Li^+ ions was higher. This evidence agrees well with previous reports stating that in the glasses with NASICON chemistry the network structure collapses around alkali ions, destroying the 'openness' of the structure and mitigating the possibility of migration for Li^+ ions [27].

As a result of heat treatment, the total activation energy decreased (0.40 eV for LATP, 0.35 eV for LATBP01 and 0.42 eV for LATBP03) whereas total conductivity increased 3–4 orders of magnitude (see Fig. 9 and Table 2) and reached the maximum value of $6 \cdot 10^{-5}$ S/cm for LATBP01 glass-ceramics.

In order to separate the bulk and grain boundary electrical responses, modelling of impedance data in the terms of the previously proposed equivalent circuits (see Fig. 7) was performed. For glasses, a simple $(P_1CR)P_2$ model was applied, in which a constant phase element

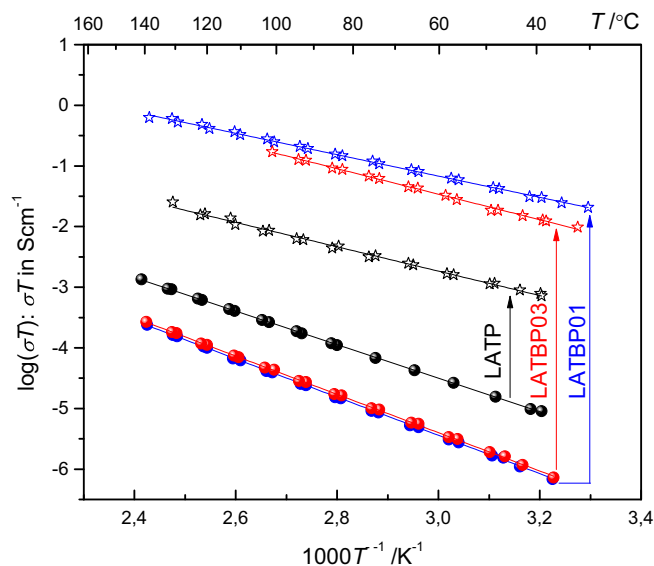


Fig. 8. Arrhenius dependence of total ionic conductivity for glasses ($\bullet$) and glass-ceramics ($\star$).

Table 2

Activation energies and room temperature (25 °C) conductivity values for investigated glasses and glass-ceramics.

Boron content	E_a (eV)	σ_{RT} (S/cm)	E_a (eV)	σ_{RT} (S/cm)
	Glasses		Glass-ceramics	
0 (LATP)	0.55	$1 \cdot 10^{-8}$	0.40	$1 \cdot 10^{-5}$
0.1 (LATBP01)	0.63	$1 \cdot 10^{-9}$	0.35	$6 \cdot 10^{-5}$
0.3 (LATBP03)	0.63	$1 \cdot 10^{-9}$	0.42	$2 \cdot 10^{-5}$

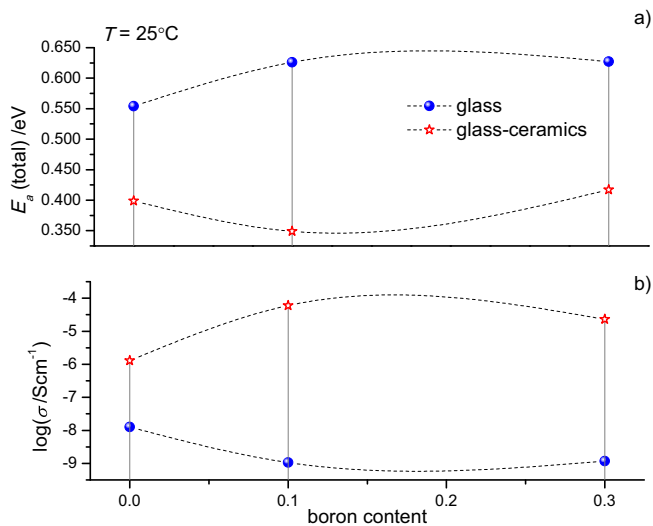


Fig. 9. Total activation energy (a) and room temperature DC conductivity (b) as a function of boron content.

P_1 and capacitor C correspond to overall relaxation processes in the bulk, resistor R represents the DC resistance of the sample and P_2 - double-layer formation at the electrodes.

For glass-ceramics, the $(C_1R_1((P_1C_2)R_2))(P_2(WC_3))$ model was applied. This circuit is similar to the one previously used in the case of typical inhomogeneous solid ion conductors [28]. The capacitor C_1 represents the geometrical capacitance of the sample. Resistors R_1 and R_2 could be associated with faster and slower transport processes, occurring within the grains (at higher frequencies) and through grain boundaries (at lower frequencies). The values of R_2 could be fitted with reasonable accuracy in the entire temperature range (from +30 up to +110 °C). R_1 could be fitted only up to +60 °C, because small

semicircle corresponding to that grain resistivity, shifted beyond available frequency limit and was practically not observed. The capacitor C_2 and constant phase element P_1 , linked in parallel to resistor R_2 correspond to electrical polarization occurring on phase boundaries (Maxwell-Wagner polarization). The electrode part of the circuit consists of the element P_2 (double layer formation) in parallel with W (Warburg diffusion) and C_3 .

Fig. 10 presents the temperature dependence of R_1 and R_2 (corresponding to bulk R_b and grain boundary R_{gb} resistance) for LATBP01 and LATBP03 glass-ceramics in Arrhenius coordinates. The activation energy for grain boundary conduction mechanism is lower in the case of LATBP01 (0.35 eV vs 0.40 eV for LATBP03). For bulk conductivity, the difference in activation energy values is much more pronounced. While E_a for LATBP01 is 0.36 eV, it reaches 0.50 eV for LATBP03. This observation could be linked with XRD refinement results - the evolution of lattice parameters indicate that same structural changes induced by boron substitution may take place in NASICON-like structure resulting in deterioration of bulk Li^+ ion transport properties. However, by taking into account Fig. 9 it may be concluded that there is a limit in this effect, as the first boron incorporation (LATBP01) decreases the total activation energy, as compared to LATP and the subsequent B-incorporation (LATBP03) leads to an increase in the activation energy.

Impedance spectroscopy studies show that although the total ionic conductivity of B-incorporated glasses is lower than for their B-free analogue, after crystallization, B-incorporated glass-ceramics exhibit better electrical properties than B-free materials. The highest conductivity value was observed for LATBP01 sample. This result may be explained by higher concentration of Li^+ ions and lower content of unwanted phases. For LATBP03, the total conductivity is lower than for LATBP01. Simultaneously, the activation energy for grain conductivity mechanism is higher.

Although the concept of boron incorporation to the LATP glass has been already explored, the present study shows more comprehensive results that may be helpful in the search of effective ways to improve the performance of LATP-based solid state electrolytes. Comparing with the previous study on similar systems done by Chen et al. in [19], it was confirmed that the glass-forming tendency of B-incorporated $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{TiO}_2-\text{P}_2\text{O}_5$ system is improved and the precipitation of unwanted phases can be restricted due to B_2O_3 addition. The crystallization of precursor glasses also resulted in the increase of ionic conductivity with respect to the glasses before heat treatment, but in the case of present study, this increase was less significant, probably because of different sintering conditions. However, the novelty of this study is the insight in the glass network of B-incorporated LATP glasses by means of Raman spectroscopy. In addition, some experimental

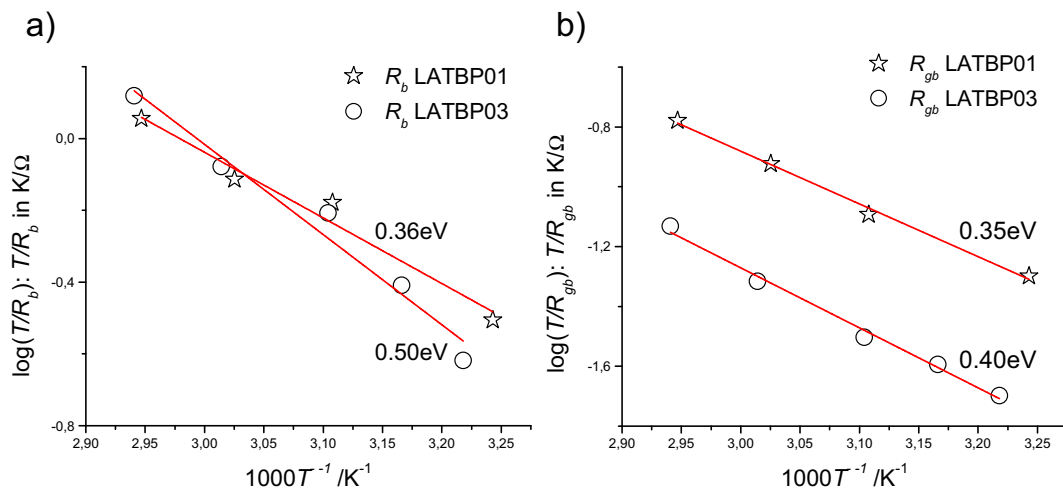


Fig. 10. Arrhenius-type dependence of fitted bulk (a) and grain boundary (b) resistance for LATBP glass-ceramics.

proofs seem to indicate that the beneficial effect of B-incorporation is limited to ca. 0.1 mol%. For higher boron contents, unfavourable structural changes in NASICON-like structure might take place, resulting in the deterioration of overall electric properties of this group of LATP-based materials.

4. Conclusions

On the basis of presented data, some conclusions concerning the impact of B₂O₃ additive on LATP-based glasses and glass-ceramics may be formulated:

- glass forming ability of Li₂O-Al₂O₃-TiO₂-P₂O₅ system could be improved by adding B₂O₃ to the melt. As evidenced by DTA studies, boron addition extends the separation zone between glass transition and crystallization.
- XRD results indicate that in B-incorporated glass-ceramics, the content of undesirable phases, in particular LiTiPO₅, is significantly reduced and the precipitation of the main, NASICON-like phase, is facilitated. Rhombohedral unit cell parameters change with boron content which may suggest that this element is incorporated into the crystal lattice and substitute for Al³⁺/Ti⁴⁺ ions.
- The structural homogeneity of the obtained LATP samples has been confirmed in Raman studies. In addition, the characteristic peaks of BO₄ and BO₃ units in B-containing LATP glasses were observed.
- impedance spectroscopy studies of LATP and LATBP materials revealed that the highest value of total ionic conductivity of ca. 6 · 10⁻⁵ S/cm was observed for LATBP01 glass-ceramics along with lower value of activation energy for grain conductivity mechanism while compared with LATBP03 material. Several factors could be responsible for this behaviour, namely: higher concentration of mobile Li⁺ ions, structural changes promoting the fast ion transport and decrease in secondary phases content.

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